# Bi-Weekly Status Report

**Project Name:** Code Sentinel – A Multi-Agent Code Review Platform
**Report Period:** 15 Oct 2026 – 31 Oct 2026
**Prepared By:** Sumit Goyal (Orchestration – on behalf of the team)
**Date Submitted:** 01/11/2026

## Project Status Summary

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|---|---|
| **Sprint/Milestone** | All five agents live; M6 mid-semester demo given on 21 and 23 Oct; the whole platform deployed to production (Render, Vercel, Grafana Cloud) at `codesentinel.scribly.fun`; M7 (Test Plan) starts |
| **% Complete (overall)** | 88% |

| Indicator | Current | Previous | Trend | Status |
|---|---|---|---|---|
| Overall Project | 88% | 72% | ↑ | **On Track** |
| Schedule | 95% | 90% | ↑ | **On Track** |
| Scope | 100% | 100% | → | **On Track** |
| Quality | 863 tests (CI green) | 688 tests (CI green) | ↑ | **On Track** |
| Team Contribution | 100% | 100% | → | **On Track** |
| Dependencies | 100% | 100% | → | **On Track** |

**Summary Notes:**
This period delivered everything planned in the last report and took the project into production. The Logic, Performance and Documentation agents joined Security and Style, so all five agents now review every pull request, reading each changed file whole. Findings from different agents on the same lines are merged into one comment, critical findings block the merge, and similar past issues are found with pgvector. The M6 demonstration was given on 21 and 23 Oct, which brings the schedule indicator back up. From 20 Oct the platform runs in production: the gateway, orchestrator and five agents as Docker services on Render, the dashboard on Vercel, custom domains, keep-warm pings, and Grafana Cloud dashboards with email alerts. The last week hardened security and usability: an audit log, signed-in devices with each device's real IP, gateway rate limits, members limited to the repositories they can open on GitHub, an optional authenticator app asked only before critical actions, live per-agent progress while a review runs, and a Next.js security patch. 77 commits from all eight members (7 to 15 each); 863 tests pass on Node 22 and 24.

## Key Accomplishments in Two Weeks

- **All five agents review every pull request** – Logic, Performance and Documentation agents on a shared LLM pass; agents read changed files whole and the Style Agent lints whole files (`34df1e3`, `fa782e5`, `e24ca67`, `d625abc`, `87fc35d`)

- **One comment per issue and merge blocking** – findings on the same lines from different agents are clubbed, with every agent's note kept; critical findings fail the Check Run (`0aacc15`, `c401c1a`, `2afd523`, `b93f067`)
- **Apply all fixes in one commit** – from the dashboard or the Check Run, with who applied them recorded (`22c1b34`, `b3ae98a`, `914adb1`, `950330c`)
- **Similar past issues with pgvector** – findings are embedded after each review and the most similar earlier issues appear in comments and on the dashboard (`0c52fab`, `484e4fc`, `56a5d64`, `4dce4eb`, `db08ad5`)
- **Branches to review per repository** – picked from the repository's real branches (`964c026`, `d411e82`, `2eec044`)
- **Security and accountability** – audit log of sensitive actions, signed-in devices with sign-out, each device's real IP behind Vercel and Render, gateway rate limits (`bad91ed`, `f89755c`, `730c9ab`, `fc7ab84`, `1b4368b`)
- **Production deployment** – Docker images on Node 22, Render blueprints for three workspaces, sleeping agents woken before a review, dashboard on Vercel, migrations at gateway start (`19a25d3`, `33c2edc`, `a73ff91`, `81a4fac`, `5be12d3`)
- **Monitoring** – Prometheus metrics on the gateway and orchestrator, Grafana Cloud dashboards and alert rules emailing the team; an alert was tested by suspending an agent (`b4edb32`, `3c5975c`, `a2751fa`, `191aaaf`)
- **Repository access and authenticator codes** – members see only the repositories they can open on GitHub; the authenticator app is optional and, once added, asked only before critical actions (`2732660`, `c6e3802`, `1d9a877`, `f12040f`)
- **Live review progress** – each agent's result is saved as it finishes, and the pull request page shows agents reporting one by one with a timer (`3117cc5`, `3c53a40`, `5f5e0b0`, `ca39a9c`)
- **Next.js security patch** – dashboard moved from 16.4.0 to 16.4.3 (`ac8f32b`)

## Planned vs. Actual

| Task/ Deliverable | Planned Completion | Current Forecast | Actual Status | Notes |
|---|---|---|---|---|
| Logic, Performance and Documentation agents | Oct 28 | — | Done (Oct 15) | All five agents run on every review, on the shared agent kit |
| Whole-file review and linked files | Oct 28 | — | Done (Oct 15) | Agents read each changed file whole; Style lints whole files |

| Task/ Deliverable | Planned Completion | Current Forecast | Actual Status | Notes |
|---|---|---|---|---|
| Separate Groq accounts per LLM agent | Oct 28 | — | Done | Each LLM agent can use its own Groq account and quota budget; in production Logic and Performance have their own, Security and Documentation share one; Gemini takes overflow |
| M6 mid-semester demo | Oct 16, Oct 23 | — | Done (Oct 21, 23) | Given in two slots on 21 and 23 Oct |
| Merge overlapping findings | Oct 28 | — | Done (Oct 16) | One comment per location, with every agent's note |
| Vector context on pgvector | Oct 28 | — | Done (Oct 17) | Gemini embeddings in Supabase; similar issues shown in comments and dashboard |
| Hosting spike on Render | Oct 28 | — | Done (Oct 20) | Went beyond a spike: the full platform is in production |
| Test Plan outline (M7) | Oct 28 | Nov 06 | Not Started | Behind deployment and hardening; scheduled for 1–6 Nov |

## Project Risks, Issues, & Dependency

| Type (ID) | Description | Impact | Owner | Mitigation/Next Step |
|---|---|---|---|---|
| Issue (I-03) | No merged work from 29 Sep to 09 Oct | Medium | Sumit Goyal | **Closed**: steady delivery since 10 Oct, all members committing each week |
| Risk (R-01) | LLM free-tier quotas shared by four LLM agents | Medium | Sumit Goyal | **Mitigated**: Logic and Performance on their own Groq accounts, quota reserved before each call, Gemini as overflow; Grafana alert when quota runs low |
| Risk (R-06) | M6 demo moved to 16 and 23 Oct | Medium | Vatsal Kamani | **Closed**: demo given on 21 and 23 Oct |

| Type (ID) | Description | Impact | Owner | Mitigation/Next Step |
|---|---|---|---|---|
| Risk (R-10) | Free hosting sleeps idle services, so the first review after a quiet spell is slow | Medium | Nevil Nandasana | Agents are woken before each review; keep-warm pings for the gateway and orchestrator; the pull request page now shows live progress while agents wake |
| Issue (I-04) | Render does not redeploy every service on its own after a push | Low | Vatsal Kamani | Services are redeployed by hand from Render after each release; checked with the health endpoints |
| Issue (I-05) | Next.js 16.4.0 had a known security vulnerability | Medium | Parin Makwana | **Closed**: upgraded to 16.4.3 on 27 Oct |
| Risk (R-11) | M7 Test Plan not started, due 06 Nov | Medium | Ayush Soni | Planned first for 1–6 Nov, built from the existing 863 tests and the live checks |
| Risk (R-09) | The requirements document is a PDF only | Low | Sumit Goyal | Changes recorded in these reports and in the System Design Document |

## Project Decisions

| Decision | Rationale | Impact | Decided By | Date |
|---|---|---|---|---|
| Three Render workspaces, slim Node 22 Docker images | Each free workspace has 750 hours a month; the slim image runs Prisma and the linters reliably | Gateway, orchestrator and agents in separate workspaces | Team | Oct. 20 |
| Agents may sleep; the orchestrator wakes them before a review | Keeps five agents within free hours | First review after idle time is slower; progress shown live | Team | Oct. 20 |
| Only the gateway and orchestrator are scraped for metrics | Scraping agents would keep them awake | Agent health and runs reported through the orchestrator | Team | Oct. 20 |

| Decision | Rationale | Impact | Decided By | Date |
|---|---|---|---|---|
| Members see only repositories they can open on GitHub | A collaborator on one repository must not see the rest of an account | Repository access recorded at each sync | Team | Oct. 23 |
| Authenticator app optional, asked only before critical actions | Mandatory codes blocked users without an app; sign-in codes slowed every visit | Codes for configuration changes, fixes, syncs and device sign-out only | Team | Oct. 23 |
| Wireframes 03, 05–08 not built as separate pages | Overview and pull request pages already cover them | Effort moved to live progress (wireframe 04) | Team | Oct. 24 |

## Project Scope Changes

| Change | Action | Reason | Schedule Impact | Approved By |
|---|---|---|---|---|
| Audit log and signed-in devices | Added | Admins can see who changed what, and users can sign out lost devices | Absorbed within the period | Team |
| Apply all fixes in one commit | Added | Accepting fixes one by one created many commits | Absorbed within the period | Team |
| Live review progress (wireframe 04) | Added | Reviews take one to two minutes on free hosting; the page now shows them moving | Absorbed within the period | Team |
| Authenticator app | Changed | From mandatory to optional, asked only before critical actions | None | Team |
| Separate review history, three-panel diff and finding-detail pages | Removed | Covered by the overview, pull request page and GitHub's own diff | Frees about one work package | Team |

## Upcoming Work (Next Two Week's Plan)

- M7 Test Plan and test-case template, due 06 Nov (Ayush Soni, coordinating; every member writes the cases for their area)
- Live end-to-end test suite against the deployed platform on the playground repository (Vatsal Kamani, Tej Prakash Tak)

- Load and failure tests: agent timeouts, quota exhaustion and cold starts (Sumit Goyal, Nevil Nandasana)
- Requirements and System Design Document updates for deployment, monitoring and access rules (Durgesh Khushlani, Parin Makwana)
- Final demonstration preparation and dashboard polish (Moksh Mehta)
- Seventh bi-weekly report (Sumit Goyal)

## Team Member Contributions

**Name:** Sumit Goyal
**Role:** Orchestration – LangGraph workflow & LLM abstraction layer

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|---|---|
| **Tasks Completed** | Design documents for whole-file reviews and pgvector; deployment view and schema diagrams; API specification of sessions, agent health and audit events; production deployment guide; recording each agent's result as it finishes for live progress; organisation picker label; demo coordination; this report |
| **Tasks In Progress** | Load and failure tests |
| **Contribution Type** | Development / Architecture / Documentation |
| **Blockers** | None |
| **Hours/Capacity** | 176/200 hrs logged |
| **Evidence** | 8 commits, e.g. `46d029b`, `532d3eb`, `1b650b8`, `3c53a40`; `services/orchestrator`, `docs/design` |

**Name:** Durgesh Khushlani
**Role:** Orchestration – Aggregation, severity ranking & Vector DB integration

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|---|---|
| **Tasks Completed** | Branches kept per repository; indexing findings after each review and finding similar past issues; service Docker images; move to Node 22; authenticator check in the stores; agent progress column for running reviews |
| **Tasks In Progress** | Requirements and design document updates |
| **Contribution Type** | Development / DevOps / Testing |
| **Blockers** | None |
| **Hours/Capacity** | 174/200 hrs logged |
| **Evidence** | 7 commits, e.g. `484e4fc`, `56a5d64`, `19a25d3`, `3117cc5`; `services/orchestrator/src/context`, `packages/db` |

**Name:** Tej Prakash Tak
**Role:** Agent Services – Security Agent & Logic Agent

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|---|---|
| **Tasks Completed** | Logic Agent; clubbing findings across agents; Apply all fixes in one commit and recording who applied them; gateway rate limits; dashboard links in fix comments; Style linters in process mode for Render; dropping the code check at sign-in |
| **Tasks In Progress** | Live end-to-end test suite |
| **Contribution Type** | Development / Security / Testing |
| **Blockers** | None |
| **Hours/Capacity** | 170/200 hrs logged |
| **Evidence** | 9 commits, e.g. `34df1e3`, `0aacc15`, `22c1b34`, `1b4368b`; `services/logic-agent`, `services/gateway` |

**Name:** Nevil Nandasana
**Role:** Agent Services – Style Agent

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|---|---|
| **Tasks Completed** | Whole-file linting and merged adjacent fixes in the Style Agent; Apply all fixes button; service metrics and the Prometheus client switch; waking agents before a review; alert counters; members limited to their own repositories |
| **Tasks In Progress** | Load and failure tests |
| **Contribution Type** | Development / DevOps / Testing |
| **Blockers** | None |
| **Hours/Capacity** | 168/200 hrs logged |
| **Evidence** | 8 commits, e.g. `87fc35d`, `b4edb32`, `a73ff91`, `c6e3802`; `services/style-agent`, `packages/service-kit` |

**Name:** Parin Makwana
**Role:** Agent Services – Performance Agent & Documentation Agent

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|---|---|
| **Tasks Completed** | Performance and Documentation agents and the shared agent LLM pass; listing and revoking sessions; recording the device per session; removing the sign-in code page; Next.js security patch |
| **Tasks In Progress** | Requirements and design document updates |
| **Contribution Type** | Development / Security / Documentation |
| **Blockers** | None |
| **Hours/Capacity** | 167/200 hrs logged |
| **Evidence** | 8 commits, e.g. `fa782e5`, `e24ca67`, `f89755c`, `ac8f32b`; `services/performance-agent`, `services/documentation-agent` |

**Name:** Ayush Soni
**Role:** Backend & Database – API Gateway & PostgreSQL

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|---|---|
| **Tasks Completed** | pgvector embedding storage and schema notes; every agent shown on a finding; audit events; browser IP forwarding and the Render proxy count; database schema diagram; step-up only for users with an authenticator; serving review progress |
| **Tasks In Progress** | M7 Test Plan and test-case template |
| **Contribution Type** | Development / Security / Documentation |
| **Blockers** | None |
| **Hours/Capacity** | 176/200 hrs logged |
| **Evidence** | 10 commits, e.g. `0c52fab`, `bad91ed`, `1d9a877`, `5f5e0b0`; `services/gateway`, `packages/contracts` |

**Name:** Vatsal Kamani
**Role:** Integrations & DevOps – Monorepo, GitHub App & Deployment

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|---|---|
| **Tasks Completed** | One comment per location; merge blocking on critical findings; similar issues in comments; repository branches from GitHub; agent health; Render blueprints and gateway start fixes; Grafana dashboards and alerts; sleeping agents shown as asleep; repository members table; M6 demo run-book |
| **Tasks In Progress** | Live end-to-end test suite |
| **Contribution Type** | Development / DevOps / Documentation |
| **Blockers** | None |
| **Hours/Capacity** | 172/200 hrs logged |
| **Evidence** | 12 commits, e.g. `c401c1a`, `2afd523`, `33c2edc`, `3c5975c`; `render`, `docs/monitoring` |

**Name:** Moksh Mehta
**Role:** Frontend – Next.js dashboard

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|---|---|
| **Tasks Completed** | New agents shown live; reviews on the overview; branch picker; seen-before issues; devices and audit log pages; agent health notices; Vercel deployment and lockfile fix; reading the gateway URL at request time; live review progress panel; M6 demo slides |
| **Tasks In Progress** | Final demonstration preparation and dashboard polish |
| **Contribution Type** | Development / Design / Documentation |
| **Blockers** | None |
| **Hours/Capacity** | 170/200 hrs logged |
| **Evidence** | 15 commits, e.g. `22bf5e8`, `0fe858c`, `81a4fac`, `ca39a9c`; `apps/dashboard` |